

Comparative analysis of classics book review data created by users across Douban and Goodreads

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Abstract

While empirical research on online book reviews has advanced our understanding of book reception and everyday reading, more cross-platform and comparative analyses are needed to cross-examine and enrich existing perspectives on these topics. This paper presents our comparative analysis of user-created reviews of “classics” across two platforms: Goodreads based in the U.S., and Douban based in China. With book tagging, rating, and reviewing data collected from the two platforms, we examined how 144 classics identified by users on Goodreads and 141 classics on Douban were characterized and received across platforms. Our analysis reveals how book reviewers’ interests in and opinions about classics differ across platforms and demonstrates the necessity of gaining a more comprehensive understanding of classics. We also contribute a parallel dataset of classics across Goodreads and Douban for future investigations. To the best of our knowledge, this is the first parallel classics dataset across Anglophone and non-Anglophone platforms. In addition, we identify three primary challenges in aligning and comparing cross-platform data regarding metadata quality, rating systems, and multi-lingual data processing. We present these challenges and discuss how we managed to overcome them. Our case study furthers existing conversations on the characterization and reception of classics online and informs future cross-platform online book review studies.

Keywords: classic books; online book reviews; comparative studies; cross-platform studies; user-generated content.

1. Introduction

Due to the lack of historical reading records, many aspects of readership studies remained theoretical or anecdotal until the 21st century (Milligan 2016; Hu *et al.*, 2023). In the last two decades, the abundance of online book reviews has provided new opportunities for empirical research on contemporary book reception, everyday reading, online reader communities, literary genres, etc. (Bartley 2009; Bourrier and Thelwall 2020; Pianzola, Rebora, and Lauer 2020; Rebora *et al.*, 2021). For instance, with book review data collected from Goodreads, a U.S.-based social reading and book reviewing website, Walsh and Antoniak empirically revealed that amateur critics have curated a narrower sense of crowdsourced classics, under the impacts of schooling, industrial stakeholders of books

(e.g. publishing, film, and television), and algorithmic moderation of e-commerce platforms such as Goodreads (Walsh and Antoniak 2021).

However, limitations and gaps also emerge from existing online book review research in digital humanities. While researchers have leveraged a variety of datasets in different languages (Maryl 2008; Zhang *et al.*, 2014; Hong *et al.*, 2017; Ehrmann and Schmale 2018; Koolena *et al.*, 2020), books being studied are mostly associated with distinguished social impacts, literary prestige, and commercial success in the Anglophone world, such as the Western canons/classics, mass market bestsellers, prize winners, etc. (Kovács and Sharkey 2014; Maity *et al.*, 2017; Ptubhof and Da 2018; Bourrier and Thelwall 2020). From these studies, reciprocal effects and multiplier

effects have already emerged where book review platforms (e.g. Goodreads, LibraryThing), booksellers (e.g. Amazon Books, Barnes and Noble), and academic book impact indexes (e.g. the MLA International Bibliography, the Open Syllabus Project) seem to construct an echo chamber by endorsing the same sets of books (Bao and Chang 2014; Maity *et al.*, 2017; Bourrier and Thelwall 2020). In contrast, other authors and their books (along with their readerships) have been historically omitted in both literary history and the publishing industry—partially due to classism, sexism, racism, colonialism, etc. (So 2020; So and Wezerek 2020). For instance, in the study of user-curated classics reviews on Goodreads, none of the 144 most shelved/tagged Goodreads classics is created by Asian, Asian American, or Indigenous authors (Walsh and Antoniak 2021). This unavoidably limits insights gained from this classic dataset, despite the size of the dataset.

In order to expand the existing research landscape, DH scholars have been actively creating more book review datasets (e.g. multi-lingual, cross-cultural, cross-genre, etc.) (Worrall 2014; Dimitrov *et al.*, 2015; Newell *et al.*, 2016; So and Wezerek 2020; Long 2021; Kotin and Koeser 2022). In particular, as Pianzola *et al.* advocated (Pianzola *et al.*, 2022), multilingual and cross-cultural datasets are much needed. Responding to such advocacy, we aim to further conversations about user-reviewed classics through a cross-platform case study. Specifically, given the Goodreads classics dataset of 144 books curated by Walsh and Antoniak (2021), we built another dataset of 141 books that were identified as “classics” on Douban, a review platform based in China.

Walsh and Antoniak’s classics list includes books that have been tagged as a “classic” the largest number of times and/or read the most as of 2019 on Goodreads. As the types of data they used (e.g. user shelving data and rich genre tags) were not available on Douban when we conducted this research, we adopted a similar but not identical approach to build our dataset. We identify Douban classics as widely reviewed Douban Top 250 Books (from 2011 to 2021) that are explicitly tagged as “classics.” We did not define the classics from any scholarly perspective; instead, we studied which books were regarded as “classics” by users on a particular platform, as Walsh and Antoniak’s strategy with Goodreads classics. Then we collected book rating data about these classics across Goodreads and Douban for comparison. As the user bases across Goodreads and Douban have significant linguistic and geographical differences, this cross-platform case study also has the potential to reveal cross-cultural perspectives on classics across primarily Anglophone and non-Anglophone readerships online.

The expected contributions of this case study are 3-fold. First, we aim to understand how book reviewers’ interests in and opinions about classics differ across platforms. For instance, what constitutes “the classics” as curated on Douban? Do Goodreads and Douban users create distinctly different lists of classics? For the classics being reviewed across linguistically and geographically distinct platforms, on which books do reviewers’ opinions diverge and converge the most? We prioritized presenting the findings regarding Douban classics and comparative analysis, as findings on Goodreads classics have been thoroughly analyzed in Walsh and Antoniak’s work. Second, regarding methodology, we pinpointed three primary challenges in aligning and comparing cross-platform data: different rating systems, varied data quality, and unique platform contexts/dynamics. Such challenges were rarely discussed in existing scholarship on online book reviews involving cross-platform datasets. We present these challenges and how we mitigate them to inform future studies. Third, we curate and contribute a parallel dataset of user-curated classics reviews across Goodreads and Douban for future investigations, which has been published on Zenodo.¹

The rest of this paper is organized as follows: Section 2 introduces the two platforms investigated and how we built the cross-platform datasets for comparative analysis. In addition, we elaborated on the challenges we confronted and the actions we took to address them. Section 3 presents our analyses and findings. Section 4 discusses our research limitations and methodological reflections. Finally, Section 5 summarizes our research contributions and implications.

2. Data sources and datasets

2.1 Data sources

The two platforms and data sources are Goodreads, based in the U.S., and Douban, based in China. Goodreads is a social reading platform devoted to online book reviews. Founded in 2007 and acquired by Amazon in 2013, Goodreads has acquired over 150 million members as of 2023 (Hu *et al.*, 2023; Stamper-Halpin 2023). While Goodreads claims to be “the world’s largest community of readers,” its active user base may not be as global as claimed: according to the analysis of a dataset of 1,872,677 Goodreads users (Sabri and Weber 2021), those based in the United States, the United Kingdom, Canada, and Australia make up 49 per cent, while users based in major Chinese-speaking geographical regions, namely Mainland China, Hong Kong, Macao, and Taiwan, only make up approximately 0.8 per cent (Hu *et al.*, 2023). In contrast, Douban’s user base is more homogeneous but complementary to Goodreads. Douban

users are based within “the national boundaries of China as well as the Greater China region” (Kiu-Chor 2007), with a primary user base in large and medium cities in mainland China (NetEase 2013). Founded in 2005, Douban has around 220 million registered members as of 2020 (Hu *et al.*, 2023).² In addition, while Goodreads focuses exclusively on books, Douban is a multi-functional platform allowing users to review a broad range of cultural items, such as books, movies, and museum exhibitions. Douban Books, among many specialized channels, is devoted to book reviews. For this paper, we used “Douban” to refer to Douban Books specifically.

To collect data about Goodreads classics, we reused the set of 144 classics identified through users’ shelving and tagging data (Walsh and Antoniak 2021). For Douban classics, we selected the 141 books that were on the Douban Top 250 Books list from 2011 to 2021 (Hu *et al.*, 2023)³ and that were tagged as “classics” (“经典” in Chinese) and/or “masterpieces” (“名著” in Chinese). At the time of our data collection, Douban only provided around ten user-generated tags on each book’s page, without specifying their selection criteria, rankings, or frequencies of usage. Table 1 shows an example of 5 Goodreads classics and 5 Douban classics. We collected data about these 144 Goodreads classics and 141 Douban classics across both platforms to create a parallel dataset.

2.2 Workflow and challenges of dataset creation

The overall workflow for building the parallel dataset consisted of three steps: (1) identifying all potentially pairable book pages across the two platforms; (2) collecting data from the selected book pages; (3) cleaning and pairing the Goodreads and Douban data to create

the parallel dataset for comparisons. While this workflow seemed straightforward, we encountered three primary challenges: (1) complexities and varied quality of online book metadata; (2) differences in information organization across the two platforms; and (3) challenges in computational processing of multilingual and cross-platform cultural data. These problems are intertwined in practice and complicated by differences in the languages of the data available on the webpages. These make it challenging to create a purely computational yet reliable workflow for webpage alignment and data collection. Therefore, we combined human inspection and manual data cleaning with computational data processing throughout our data collection and analysis.

First, book metadata is complex in the real world. Books under the same titles and by the same authors are not necessarily different copies of the same work; they might be different volumes of the same multi-volume work. Meanwhile, the same work might be published under different book titles and varied author names after translation. For instance, *Animal Farm* in English by George Orwell has been renamed as 动物农场, 动物庄园, and 动物农庄 when published in Chinese.⁴ Therefore, computational metadata matching can be inaccurate for pairing books across platforms and languages. Multiple factors should be considered when verifying whether multiple books are different editions of the same work. In addition, book data quality varies. For instance, we constantly found incorrect and missing metadata on Goodreads and Douban webpages. Figure 1 shows two exemplary webpages with metadata errors. Figure 1a presents a Douban book page screenshot where metadata for *E.T.* was mistakenly shown on the book page for *A Dog of Flanders* (Douban: 弗兰德斯狗 2024); Fig. 1b presents a

Table 1. An overview of Goodreads classics and Douban classics.

	Number of books	References	Five platform-based classics
Goodreads	144	Walsh and Antoniak (2021)	• <i>The Iliad</i> by Homer • <i>Beowulf</i> by Unknown • <i>The Canterbury Tales</i> by Geoffrey Chaucer • <i>The Prince</i> by Niccolò Machiavelli • <i>Romeo and Juliet</i> by William Shakespeare
Douban (books)	141	Hu <i>et al.</i> (2023)	• <i>To Live: A Novel</i> (original title in Chinese: 活着) by Yu Hua (余华) • <i>Fortress Besieged</i> (original title in Chinese: 围城) by Qian Zhongshu (钱钟书) • <i>The Story of the Stone</i> (original title in Chinese: 红楼梦) by Cao Xueqin(曹雪芹) • <i>The Dancing Girl of Izu and Other Stories</i> (original title in Japanese: 伊豆の踊子) by Yasunari Kawabata (川端康成) • <i>One Piece</i> (original title in Japanese: ワンピース) by Eiichiro Oda (尾田栄一郎)



Figure 1. Screenshots of Douban page for *A Dog of Flanders* (a) and Goodreads page for 珞珈野火集 (b), both pages accessed on 11 July 2023.

Goodreads page where a summary of the book 我所理解的世界 by 闫丘露薇 was shown on the page for the book 珞珈野火集 by 刘道玉 (Goodreads: 珞珈野火集 2024). Such erroneous information makes manual inspection necessary for data quality control.

Second, neither Douban nor Goodreads has comprehensive bibliographic control, which limits what web pages and data can be retrieved for comparisons. This problem, when combined with the cross-platform differences in data organization (explained in the next paragraph), poses challenges to (1) aggregating data about the same work platform-wise; and (2) aligning data about different versions/editions of the same works across platforms.

Differences in data organization refer to how book ratings and reviews are managed differently across platforms. On Goodreads, if a book has been linked with other editions/versions of the same intellectual work, the ratings shown on its book page are most

likely based on aggregated ratings of “all editions” of the same work, instead of the ratings for the specific editions/versions of the works presented on the webpage. Figure 2 shows an example of the book page for the most-rated edition of *Jane Eyre* (Goodreads: *Jane Eyre* 2024). The average ratings, number of ratings, and number of reviews were all based on “all editions.”⁵ To find the rating for this particular book, one needed to click on the rating details to find it in the hover box (4.13/5). However, the distributions of the 1-to-5-star ratings for a particular edition remain unavailable. Therefore, to collect both average and distributed ratings on Goodreads, we chose to use the “all editions” data instead of the “this edition” data, since the “this edition” ratings were only partially available.

In contrast, book ratings on Douban’s book pages vary individually by edition, while “all editions” ratings are unavailable. For instance, Fig. 3 shows

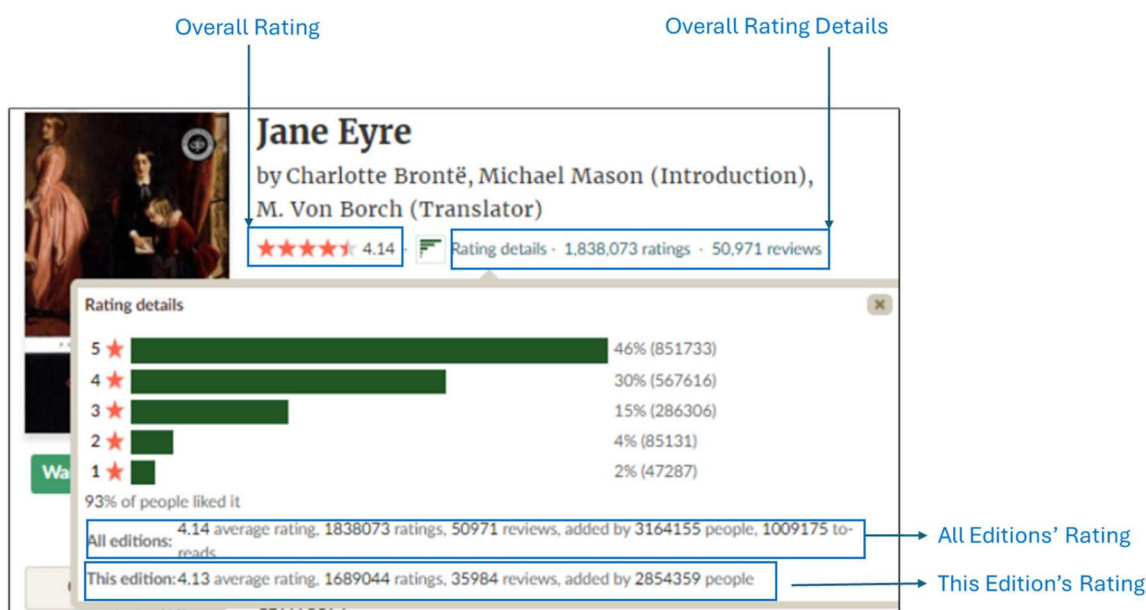


Figure 2. Goodreads page for *Jane Eyre*, with a hover box showing detailed ratings for “All editions” and “This edition,” data accessed in April 2022.

different editions of *Jane Eyre* on Douban.⁶ Each edition has its unique rating and total number of ratings. Given (1) the differences in data organization and rating availability across platforms; and (2) the fact that neither Goodreads nor Douban can correctly and comprehensively identify and aggregate all different versions/editions of the same works, it was impossible to align or compare the “complete sets” of the same works across the two platforms. Therefore, our data collection was narrowed down to comparing data on the most-rated/most-reviewed book pages of the same work across Douban and Goodreads.⁷ Additionally, the distributions of ratings used for comparison were measured by percentages of 1-to-5-star ratings instead of absolute numbers of ratings, to reduce the negative impacts of differences in total numbers of ratings across platforms.

Third, as the data collected from Douban and Goodreads were in different languages, there are other challenges for computational data processing. For instance, processing named entities in Chinese remains a challenge in existing natural language processing research (Cheng *et al.*, 2011; Peng *et al.*, 2015). Some named entities in Chinese we collected were encoded with different Chinese text encoding standards (e.g. Unicode, GB2312, and GBK). This means that while the named entities look the same to human eyes and indeed refer to the same entities, they register as different entities to computers, due to their encoding differences. Such problems make additional manual inspection necessary.

Furthermore, for Douban classics that were lesser known and rarely reviewed on Goodreads, their search results varied by the query term and language used. Our search with the book 第一炉香 (English title as *Aloeswood Incense*) by the author 张爱玲 (English name: Eileen Chang) demonstrated this point. To find all potential matches for this book on Goodreads, we searched for its book title 第一炉香 in simplified Chinese, 第一爐香 in traditional Chinese, and *Aloeswood Incense* in English. As a result, 0, 1, and 1 books were returned, respectively. Given the fact that different queries might lead to different items retrieved, we manually searched for each Douban classic’s original title (based on the first publication of the work) and its most popular title in English (if it exists) to identify potential matches available on Goodreads. In most cases, at least one of these queries returned the most rated book page. For the exceptions, we searched for the book with book titles in other languages, book author name(s), and book series name, to find the potential best matches.

Finally, based on these cross-platform matches (paired URLs), we collected data including book meta-data, ratings, and user-generated tags, among others, associated with the 141 Douban classics and 144 Goodreads classics. Due to the challenges discussed, we combined manual and computational approaches to collect and align cross-platform data for comparisons. In addition, we manually added “language of the



Figure 3. Ratings of different Chinese editions of *Jane Eyre*, with the book title in simplified Chinese as 简爱, on Douban, accessed in April 2022.

first publication” tags to the books to note their cultural backgrounds. We used the language of the first publication tags as they indicate the earliest and primary readers of the book. We did not use tags such as the authors’ motherland or country of citizenship as our research features the cultural backgrounds of the readership instead of authorship.

3. Data analysis and findings

3.1 Features of Douban classics

First, to understand “what counts as a classic” on Douban, we analyzed the 141 Douban classics with regard to the language of first publication and book genre. Unlike the Goodreads classics, which include “no works by Asian, Asian American, or Indigenous

authors,” Douban classics includes 73 books (52 per cent) that were first published in Chinese (41 per cent) and Japanese (11 per cent). The rest of the classics were first published in English (26 per cent), French (9 per cent), German (4 per cent), Russian (2 per cent), Bengali (2 per cent), Arabic (1 per cent), Spanish (1 per cent), Italian (1 per cent), Greek (1 per cent), and Danish (1 per cent), as is visualized in Fig. 4. This suggests that classics curated on Douban involve works from various linguistic and geographical backgrounds, and are significantly different from Goodreads classics which are primarily published in white western languages.

Second, we analyzed the genres of Douban classics. The genre tag for each book was primarily selected from the tags on its web page.⁸ For a few books whose genre-specific tags were unavailable (e.g. all their Douban tags were about plots or topics), we examined the Goodreads genre tags for the same work to find a genre tag. We acknowledge the complexity and fluidity of genres, and we understand that genres are neither mutually exclusive nor collectively comprehensive. We only picked one genre tag for each book to (1) reflect the most prominent understanding of genres on Douban; and (2) streamline our book genre analysis. A more inclusive and nuanced analysis of all book tags, including more genre tags, is presented in Section 3.3.

Figure 5, the sunburst chart, displays distributions of books by genre in a hierarchical manner: The inner ring represents the top of the hierarchy, fiction (in blue) or nonfiction (in orange); the outer ring shows

how the fiction and nonfiction works are broken into more specific genres, with numbers after the genres indicating how many works belong to this group. For the genres that only have one work associated with them, their genre information and number of works were listed right next to their corresponding parts for the sake of space. For instance, “philosophy 7” in the orange outer ring indicates that seven nonfiction Douban classics are tagged as “philosophy,” whereas “absurdism 1” next to the outer blue ring indicates that one fiction Douban classic is tagged as “absurdism.”

Overall, 121 of the 141 Douban classics (86 per cent) are fiction works, while the remaining 20 are nonfiction. Fiction works include prose, poetry, prose poetry, short stories, novellas, and novels; nonfiction works range from literary criticism, dictionaries, and monographs in various disciplines (e.g. politics, economics, film, and art). According to Fig. 8, historical fiction is the largest genre of classics, taking up 18 per cent of all the Douban classics. Historical fiction is also the most prominent group of classics on Goodreads. Romance and manga are the second most popular genres, each taking up 11 per cent of the total. Such findings suggest the significance of genre fiction among Douban classics, which is similar to the case of Goodreads classics.

One of Douban classics’ most significant differences from Goodreads classics is the number of non-fiction works. Goodreads classics only have seven non-fiction works, including one book on political philosophy

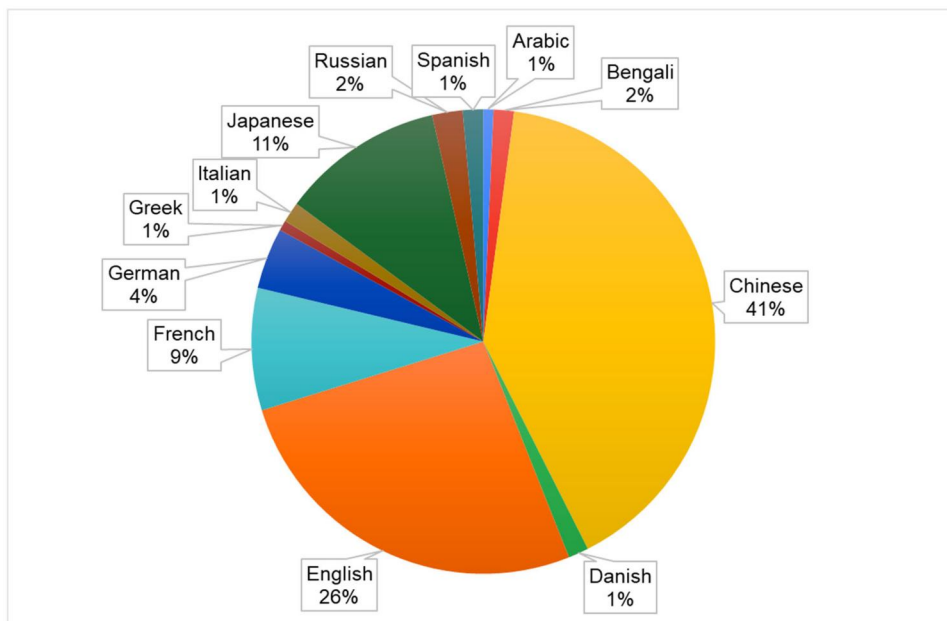


Figure 4. Percentages of Douban classics by the language of first publication.

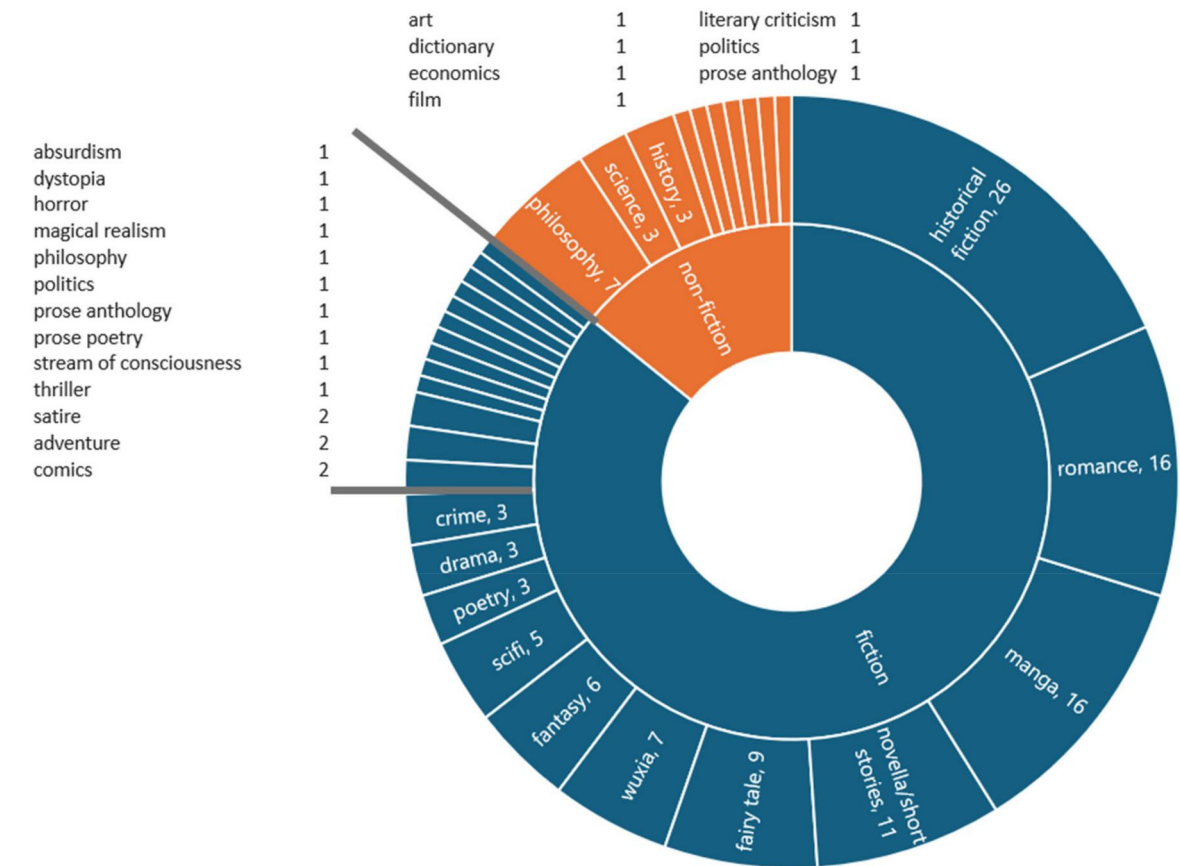


Figure 5. Number of Douban Classics by genre.

(*The Prince*), one self-help book (*As a Man Thinketh*), one true crime book (*In Cold Blood*), three memoirs/autobiographical diaries (*Narrative of the Life of Frederick Douglass*, *Tuesdays with Morrie*, and *The Diary of a Young Girl*), and one extended essay (*A Room of One's Own*). Douban classics include twenty nonfiction works. Their sub-genres are also more diverse, as is shown in Fig. 5. Another major difference is the number of manga,⁹ sixteen of the 141 Douban classics are manga works while none of the 144 Goodreads classics is. Compared to Goodreads users, Douban users' unique interest in/familiarity with Japanese manga might be partially explained by the "similar cultural-economic experience" shared by the Asian audience, and "the possibility of modernity in an Asian image" provided by Japanese manga in the 20th centuries (Wong 2010).

Compared to reviews for fiction works, Douban non-fiction classics' reviews tend to reflect important sociocultural changes in China. For instance, many reviews on the Douban page for *Xinhua Dictionary* (in Chinese: 新华字典), the most influential Chinese-language dictionary, touched on the new Internet

buzzwords that had been added, and annotations that had been modified due to social changes. There are also many nostalgic reviews on the webpage for another non-fiction classic, *The Encyclopedia for Chinese Young Adults and Children* (in Chinese: 中国少年儿童百科全书). This set of four books was first published in 1991 as the first of its kind since the foundation of the People's Republic of China, with unprecedented investment, editing work, and media coverage. Many reviews recalled how the Encyclopedia opened a new world of science for their generations of young readers and expressed how grateful they were for their families' generosity in spending half of the household's monthly income on this set of books.¹⁰ Such unique historical and sociocultural backgrounds associated with these non-fiction books' popularity in China partially explain why these books were not as popular on Goodreads.

3.2 Douban classics in comparison with Goodreads classics and Chinese school syllabi

We compared Douban classics with Goodreads classics and Chinese school syllabi, respectively. Only twenty-

Table 2. The twenty-three Douban-Goodreads shared classics, with book metadata collected from Goodreads and Douban.

Book title	Language of first publication	Year of publication (based on Goodreads data)	Douban rating	Goodreads rating
<i>Hamlet</i>	English	1603	4.3	4.03
<i>Robinson Crusoe</i>	English	1719	4.05	3.68
<i>Pride and Prejudice</i>	English	1813	4.45	4.27
<i>Oliver Twist</i>	English	1838	4.05	3.87
<i>Jane Eyre</i>	English	1847	4.25	4.13
<i>Wuthering Heights</i>	English	1847	4.2	3.87
<i>Peter Pan</i>	English	1904	4.4	4.02
<i>The Great Gatsby</i>	English	1925	4.2	3.93
<i>Brave New World</i>	English	1932	4.55	3.99
<i>Gone with the Wind</i>	English	1936	4.65	4.3
<i>The Old Man and the Sea</i>	English	1952	4.2	3.79
<i>The Return of the King</i>	English	1955	4.5	4.54
<i>Lolita</i>	English	1955	3.85	3.88
<i>To Kill a Mockingbird</i>	English	1960	4.6	4.27
<i>The Hunchback of Notre-Dame</i>	French	1831	4.2	4
<i>The Count of Monte Cristo</i>	French	1846	4.5	4.26
<i>Les Misérables</i>	French	1862	4.5	4.18
<i>The Little Prince</i>	English and French	1943	4.5	4.31
<i>The Metamorphosis</i>	German	1915	4.2	3.83
<i>Chess Story</i>	German	1942	4.55	4.27
<i>Crime and Punishment</i>	Russian	1866	4.6	4.23
<i>Anna Karenina</i>	Russian	1878	4.6	4.06
<i>One Hundred Years of Solitude</i>	Spanish	1967	4.65	4.05

three Douban classics are simultaneously Goodreads classics. Table 2 presents more information about these books: fourteen were first published in English, three in French, one in both English and French, two in Russian, two in German, and one in Spanish. None of these books was first published in Chinese, or any other Asian languages. These classics are also relatively “old.” The earliest work is *Hamlet* published in 1603, and the most recent work is *One Hundred Years of Solitude* published in 1967. Most Douban-Goodreads shared classics are well-known classics in Western contexts, whereas none of them is regarded Asian or African literature. These books’ average ratings are 4.37 on Douban and 4.08 on Goodreads; Douban ratings are higher than their Goodreads ratings, except for *Lolita*. This suggests that Douban reviewers might speak of these classics more highly than Goodreads reviewers.

Next, we compared Douban classics with school syllabi in China. Existing research found that on Goodreads, “More than a third of the Goodreads classics authors are specifically recommended by the AP English program, and about half rank within the top 200 most assigned college-level authors” (Walsh and Antoniak 2021). Such findings suggest that schooling has a significant impact on the Goodreads classics. To verify if this is the case with Douban, we compared the Douban classics with the books listed in prominent references for school education on Chinese (as a course

subject similar to English courses in the West) in China: (1) The Reading Guidance for Primary and Secondary School Students from the Basic Education Curriculum and Textbook Development Center of the Ministry of Education of China (2020 Edition)¹¹; (2) The 100 Must-Read Books Recommended by The Ministry of Education for College Students¹²; and (3) bibliographies in the Chinese Curriculum Standards for Compulsory Education (2011 Edition), the General High School Chinese Curriculum Standards (2017 Edition), and the Introduction to Classical Books in “Ministry Edition” Chinese textbooks.¹³

Among the 141 Douban classics, thirty-eight books (27 per cent) are on at least one of these school syllabi book lists. Seven of these books are also Goodreads classics¹⁴ and twenty-two works were first published in Chinese. Notably, three authors had multiple works that were both (1) regarded as classics on Douban and (2) recommended by the school syllabi: Lu Xun,¹⁵ Lao She,¹⁶ and Rabindranath Tagore.¹⁷ In contrast, some authors had multiple works regarded as classics on Douban, but few were listed in the syllabi. For instance, popular Chinese fiction writers Jin Yong¹⁸ and Eileen Chang¹⁹ each has six works that were curated as classics on Douban; among these twelve books, only one of Jin Yong’s works is listed in the school syllabi. These findings suggest that Douban has a unique collection of classics books which diverges from the

school syllabi in certain aspects (e.g. coverage of popular WuXia fiction and romance novella).

3.3 Reception of classics across Goodreads and Douban

We compared the reception of user-curated classics across Goodreads and Douban, by genre tag, book rating, star-wise distribution of ratings, number of ratings, and number of book reviews. More specifically, we compared data across platforms by group: Douban exclusive classics, Goodreads exclusive classics, and Douban-Goodreads shared classics.

Figure 6 provides a graphic overview of the genre tags used across Goodreads and Douban for describing and organizing the classics. The original tags collected from Douban were in Chinese. We translated the tags into English with Google Translate and manually refined the translations.²⁰ In the six tag clouds, the larger the text, the more frequently the tags have been used; different colors are used only to visually distinguish the keywords. According to Fig. 6, words such as “literature,” “fiction,” and “classic” are frequently used on both platforms for tagging all groups of classics, while certain tags are significantly more popular on one platform over the other, such as frequent usage of “gothic” and “historical fiction” on Goodreads. Tags like “martial arts” and “manga” were exclusively used for tagging the Douban exclusive classics, since only Douban classics included such works. Platform-wise, classics are often tagged with “academic” and “school” on Goodreads, while none of the Douban tags for classics reflected the impact of schooling or academia. This speaks to our findings in Section 3.2 that Douban classics seem to be less impacted by schooling compared to Goodreads. Another notable finding is that most of the shared Douban-Goodreads classics were Anglophone works and are highlighted as “foreign literature” by the tags. Such differences in book lists and tags indicate high-level divergences of reading interest and tagging behavior in classics across platforms.

We also noticed some nuanced differences in genre tags used across platforms. For instance, there are seven WuXia works regarded as classics on Douban. WuXia in Chinese is composed of two Chinese characters that mean “martial arts” (“Wu,” in Chinese: “武”) and “chivalry and benevolence” (“Xia,” in Chinese “侠”) respectively. It is a genre fiction type that typically features the growth and adventures, both physical and mental, of martial artists in ancient China. On Goodreads, the seven WuXia Douban classics are tagged with a combination of “historical fiction,” “adventure,” “combat,” and “martial arts,” which seem to cover features of “Wu” but miss the unique spiritual contexts of “Xia.” On Douban,

“WuXia” is a distinctive genre. Popular Goodreads tags for WuXia, such as “historical fiction,” are rarely used for the works on Douban. Such differences most likely reflect a divergent understanding of WuXia as a genre across platforms, highlighting the distinctive appreciation and portrayal of martial arts’ spirits within broader contexts of Chinese literature and culture.

In terms of cross-platform ratings, Fig. 7 visualizes the rating distributions of the three groups of classics. In general, the ratings of Goodreads and Douban classics were concentrated in the range of 3.8 to 4.7 (out of 5); within this range, Goodreads ratings are relatively lower than Douban’s, as the aforementioned case of Douban-Goodreads shared classics shows (see Table 2). We further examined the difference in ratings by group: (1) Douban exclusive classics; (2) Goodreads exclusive classics; and (3) Douban-Goodreads shared classics. Figure 8 provides the descriptive statistics by group where we quantified the differences in overall ratings across platforms by subtracting the Goodreads ratings from their paired Douban ratings. It confirms that across classic groups, Douban ratings are higher than Goodreads ratings.

Such differences in overall ratings might be partially explained by the total number of ratings across platforms, as presented in Fig. 9. On average, a classic received about 440,000 more ratings on Goodreads. Most classics have received more ratings on Goodreads than Douban, except for the Douban exclusive classics that were mostly first published in Chinese and Japanese. Such a larger volume of ratings suggests more possibly divergent opinions from broader reader communities, which contributes to lower ratings on average. Another factor to be considered is the ratings we used for comparison: Goodreads ratings are mostly aggregated (from various editions of the same work), while Douban ratings are based on the most rated single edition of the classics. The more editions’ data included, the more likely we are to find larger rating divergences.

In addition to comparing the overall ratings, we analyzed the differences between the parallel distributions of 1-star to 5-star ratings across platforms. Through the example of *Who Moved My Cheese*, Fig. 10 explains why we considered star-wise distributions of ratings important for rating comparisons.²¹ On Douban, as shown in Fig. 10a, this book got a 7.6/10 (equivalent to 3.8/5, as both platforms use the weighted means of the 1-to-5-star ratings as their “average ratings”), however, only 26.4 per cent were 5-star reviews. Most Douban raters rated it 4-star (39.9 per cent) and 3-star (27.8 per cent). On Goodreads, as shown in Fig. 10b, this book received a 3.86/5 rating with 35 per cent of 5-star ratings and 30 per cent of 4-star ratings. While the average ratings



were close (a difference of 0.06 on a 1-to-5 scale), their different percentages of star-wise ratings suggested that readers across two platforms evaluated the book

differently. This exemplifies the importance of examining the distribution of ratings in cross-platform rating comparisons.

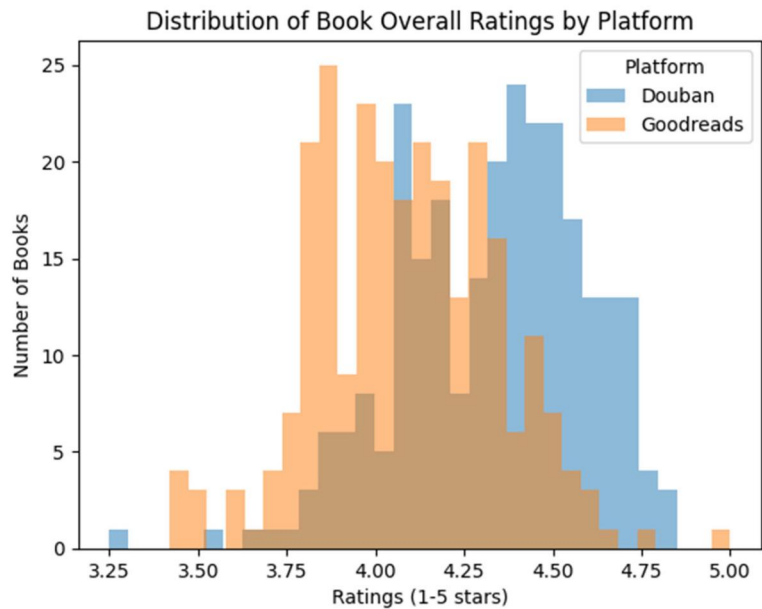


Figure 7. Distributions of book ratings of the Goodreads and Douban classics.

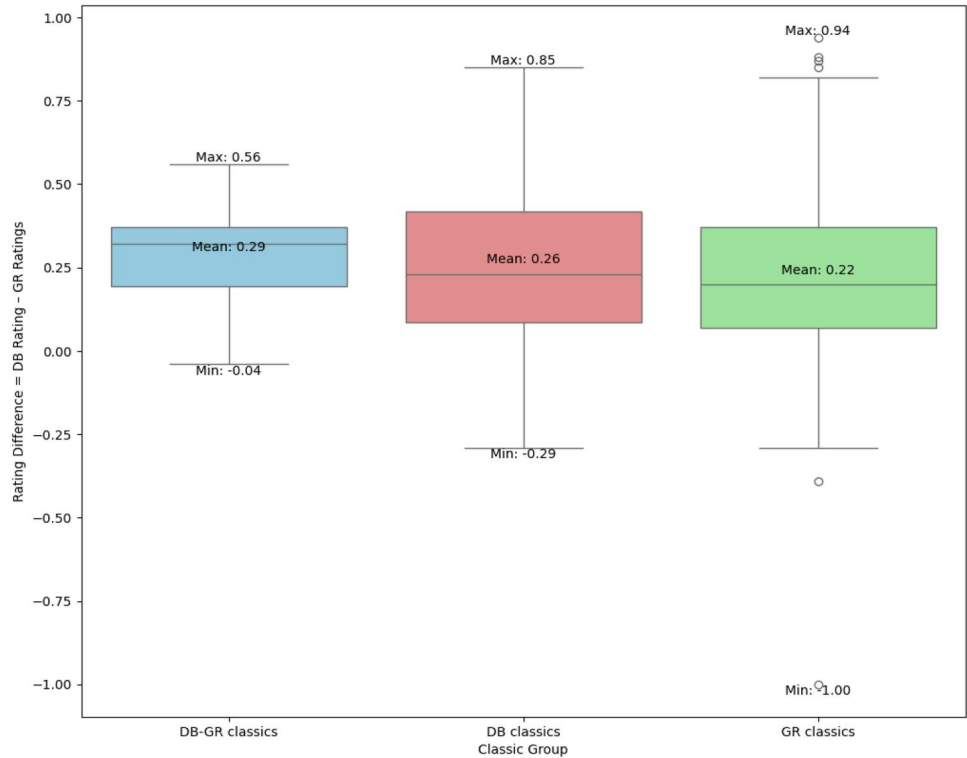


Figure 8. Differences in overall ratings by classic group.

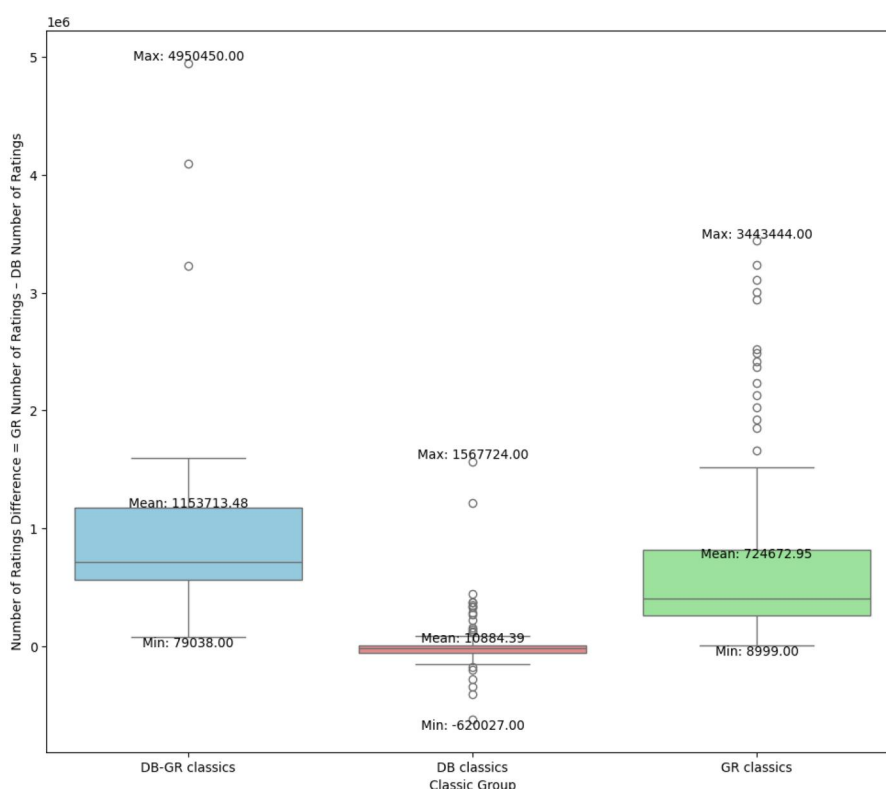


Figure 9. Differences in total numbers of ratings by classic group.

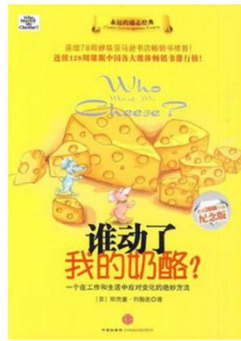
In contrast, [Fig. 11](#) shows a case where both overall ratings and star-wise distributions of the ratings are close across Goodreads and Douban: *A Global History: From Prehistory to the 21st Century* by Leften Stavros Stavrianos received high overall ratings on Douban (9.1 out of 10, equivalent to 4.55 out of 5)²² and Goodreads (4.60 out of 5), as shown in [Fig. 11](#), with similar percentages of 1-to-5-star ratings. In contrast, star-wise distributions of ratings diverged significantly on the Chinese and Japanese works that were rarely rated/reviewed on Goodreads, such as the aforementioned *Xin Hua Dictionary* which received over 7,000 ratings on Douban and only 14 on Goodreads as of May 2024.

We quantified the divergences of distributions of ratings with Jensen-Shannon Divergence (JSD), which measures the similarity between two distributions based on the Kullback–Leibler divergence (KLD). Kullback–Leibler divergence measures differences between two sets of values by their relative magnitude. It has been leveraged in cultural analytics ([Chang and DeDeo 2020](#); [Underwood and So 2021](#)), with an edge in capturing the difference in “significance” of changes and differentiating smaller changes from larger ones

([Camargo et al., 2021](#)). However, KLD is asymmetric. Therefore, we used JSD, which is symmetric with a finite value, for our analysis. Specifically, we measured JSD with the Jensen-Shannon distance, the square root of the JSD, provided by SciPy library.²³ We found the JSD values of cross-platform distributions of star-wise ratings close across the three groups of classics, with similar mean Jensen-Shannon distances across the three groups: 0.200 for Douban exclusive classics, 0.201 for Goodreads exclusive classics, and 0.208 for the Douban-Goodreads shared classics.

Finally, we compared the total number of reviews these classics received across platforms. As shown in [Fig. 12](#), Goodreads exclusive classics and Douban-Goodreads shared classics received many more reviews on Goodreads compared to Douban. On average, Goodreads classics received over 16,000 more reviews than their Goodreads matches, and Douban-Goodreads shared classics received around 6,000 more reviews than Douban. The largest positive difference lay in the total numbers of reviews for *The Alchemist* by Paulo Coelho: it received 92,829 reviews on Goodreads and only 2,970 reviews on Douban. However, Douban exclusive classics received fewer

谁动了我的奶酪？



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作者: [美] 斯宾塞·约翰逊
出版社: 中信出版社
原作名: Who Moved My Cheese?: An Amazing Way to Deal with Change in Your Work and in Your Life
译者: 魏平
出版年: 2010-5
页数: 105
定价: 22.00元
装帧: 精装
ISBN: 9787508619439



(a)



(b)

Figure 10. *Who Moved My Cheese's* rating pages on Douban (a) and Goodreads (b), both web pages accessed on 6 May 2024.

reviews on Goodreads. On average, each Douban exclusive classic received 9,512 more reviews on Goodreads than those exclusively on Goodreads. The largest negative difference lay in the total number of reviews for *To Live* (in Chinese: 活着) by Yu Hua: this book received 169,574 reviews on Douban but only 1,343 reviews on Goodreads.

4. Discussions and conclusions

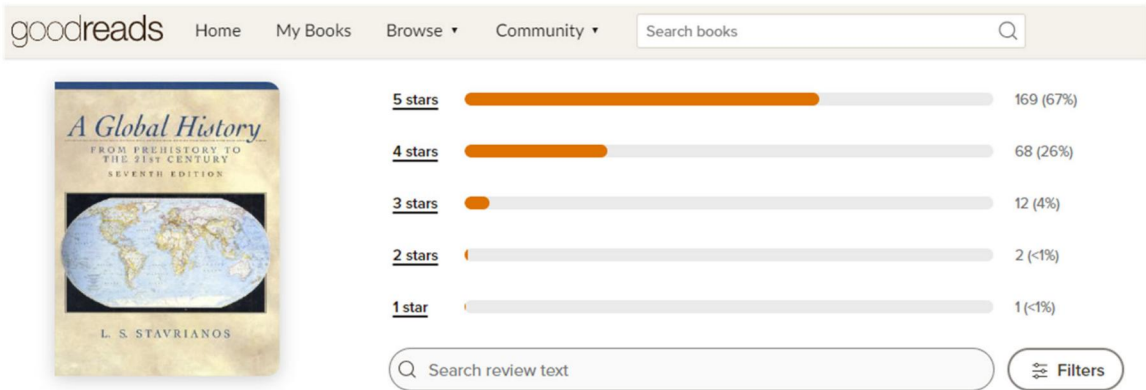
By creating a parallel dataset about (1) 141 classics curated on Douban, based in China and (2) 144 classics curated on Goodreads, based in the U.S., we empirically compare and analyze the characterization and reception of classics across primarily Anglophone and non-Anglophone social reading platforms. Our work contributes (1) a classic dataset for comparative analysis and (2) a case study of cross-platform differences in

the reviewing of “classic” books. They serve as preliminary steps toward multi-platform and comparative analyses of online book reviews which can provide a more comprehensive understanding of literature reception.

Our case study has three primary findings. First, Douban classics are significantly different from those on Goodreads. While the 141 Goodreads classics include no works by Asian authors, over half of Douban classics were written by Asian authors. Only 23 books were regarded as classics on both Goodreads and Douban. Almost all of these cross-platform classics are American and European literature works. Second, unlike Goodreads where schooling plays a significant role in characterizing classics, Douban classics diverge from the reading lists recommended in school education. For instance, Douban classics feature genre fiction and other books that are rarely seen in Chinese



(a)



(b)

Figure 11. *A Global History: From Prehistory to the 21st Century*'s rating pages on Douban (a) and Goodreads (b), both web pages accessed on 18 May 2024.

school syllabi, such as Chinese WuXia novels and Japanese manga. Popular Goodreads tags for classics such as “Academic > School” and “Read for School” were not seen in Douban tags for classics. Third, in terms of rating differences for all classics, book ratings diverge the most on the Douban exclusive classics which are lesser-known and rarely rated on Goodreads.

These findings reflect cross-platform differences in reading interests and characterization of classics. They suggest the importance of broadening the existing on-line book review research scope to acquire a more comprehensive understanding of classics online. They also demonstrate how cross-platform and comparative analysis can help us reflect on existing views about the

development of literary genres and the reception of books, as findings might vary significantly and individually across platforms and readerships.

Methodologically, we identified three primary challenges in creating parallel datasets across platforms for meaningful comparisons: (1) the complexities and varied quality inherent in online book metadata; (2) different information organization systems across platforms; and (3) the potential pitfalls of computational processing of multilingual cultural data. We also discussed the actions taken to mitigate these challenges. Our parallel datasets and associated workflows are unavoidably limited by the data available on each platform and their pairable items. Nevertheless, we want to highlight a few questions for future cross-

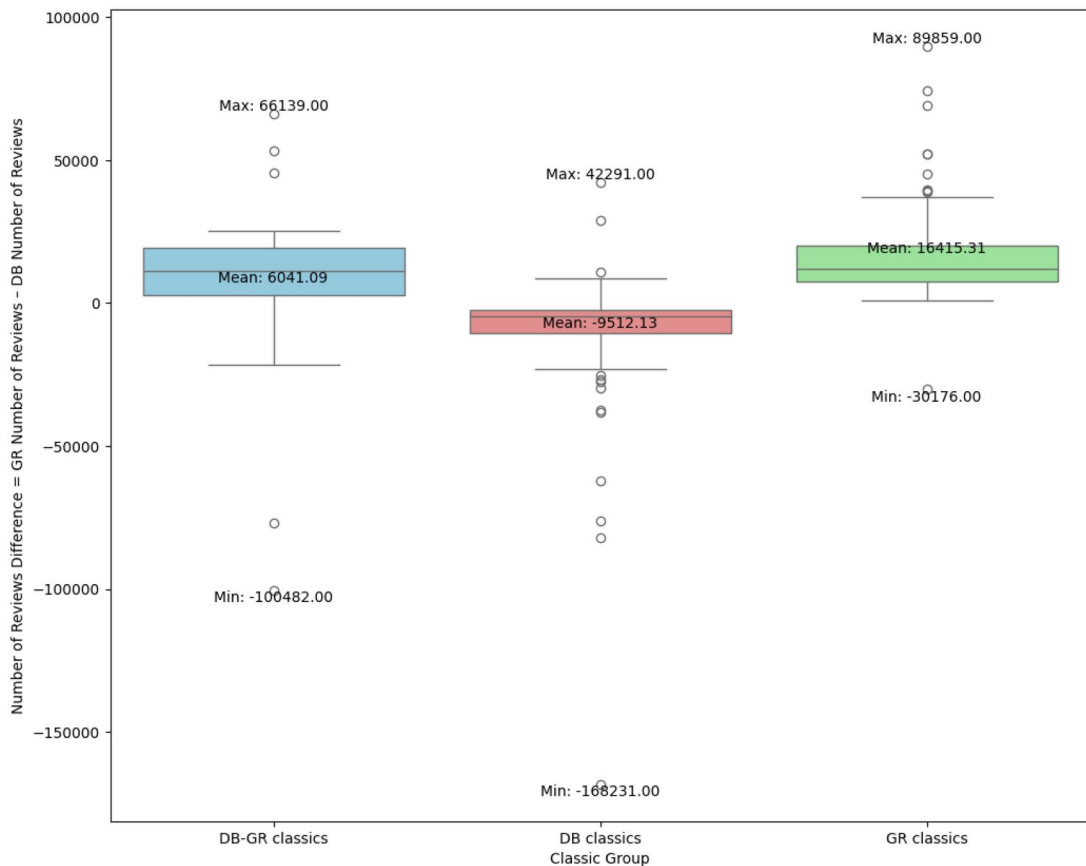


Figure 12. Differences in the numbers of reviews across classics groups.

cultural and comparative studies that emerged from our exploratory work: To what extent are data acquired from different platforms aligned for comparison? What are the unique features of each platform's datasets and what are the potential gaps across platforms? If apples-to-apples comparisons are not practical given the cross-platform differences, how can researchers make reliable use of the limited data? Furthermore, to further cross-platform studies that involve data (1) collected from linguistically, geographically, and/or culturally varying user bases; and (2) managed by different systems/platforms, future research should consider incorporating platform studies and sociotechnical system theories for more contextualized and critical data analyses.

Moving forward, we aspire to expand our comparative analysis to other platforms to cross-examine our findings with Goodreads and Douban. We are collaborating with colleagues who work with online book reviews in other non-Anglophone languages on building multilingual and parallel book review datasets to further diversify research data sources and perspectives. We also intend to investigate some of

the contributing factors to cross-platform differences. For instance, qualitative scholarship and Douban reader discussions have argued for the impact of paratexts and translations in the reception of two Douban-Goodreads shared classics, *Lolita*, and *One Hundred Years of Solitude*, in China (Bai 2018; Santana-Acuña 2020). We look forward to combining computational methods and book history/bibliography studies to (1) analyze data about different editions of the same classics; and (2) reveal the roles of paratexts and translations in the “coronation” of cross-cultural classics.

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Notes

1. Please see the User-Selected Classic Books across Douban and Goodreads Dataset on Zenodo, available at <https://doi.org/10.5281/zenodo.15252768>

2. Although Douban claims more registered users than Goodreads, Douban Books is only one division of the whole platform. Not all Douban users were necessarily active Douban Books reviewers.
3. Douban Top 250 Books has been regarded as the most popular and impactful book list on Douban. We built a dataset of the Douban Top 250 Books list from 2011 to 2021 in our prior work and reused the list to select books to investigate for this case study.
4. These three Chinese translations generally mean the same thing, “animal” (动物 in Chinese) and “farm” (农场/庄园/农庄 in Chinese). Regarding the nuanced differences in Chinese translations of “farm,” “农庄” emphasizes agriculture and farming activities, while “庄园” has a flavor of private property owned by a richer/higher class. “农场” is the most common Chinese translation of “farm.”
5. As aforementioned, due to limitations in the platforms’ bibliographic control and data quality, the “all editions” data on Goodreads are not necessarily reliable or comprehensive. According to our investigations, they might include pages for a different work and/or exclude relevant pages for the same work.
6. We searched for the title in simplified Chinese on Douban. The returned books shown in the screenshot are simplified Chinese editions of the book.
7. For each work, we collected and compared data about its (1) most-rated/reviewed page on Douban and (2) most-rated/reviewed page on Goodreads. The Douban data collected from the selected page is based on a single edition of the work, while the Goodreads data on the page is often based on multiple editions of the same work.
8. The tags available on online book review platforms are often added by users and moderated by the platforms’ algorithms. These tags provide various information, such as literary genre, descriptive book metadata, and user activities (e.g. tagging the books as “to read” and “unfinished” works). Such tags have been scraped from book web pages and broadly studied for illuminating folksonomy and social tagging in scholarly research. The tags used for our research were collected from the Douban front pages in 2022; such tags are no longer available on the pages as we draft this paper in 2024.
9. Manga is generally defined as comics and graphic novels from Japan.
10. The reviews are available on the Douban book page: <https://book.douban.com/subject/1028409/>.
11. Document title in Chinese: 教育部基础教育课程教材发展中心 中小学学生阅读指导目录 (2020 年版), accessed from http://www.moe.gov.cn/jyb_xwfb/gzdt_gzdt/s5987/202004/W020200422556593462993.pdf.
12. Document title in Chinese: 教育部推荐大学生必读书目100本, accessed from <https://lib.ecjtu.edu.cn/info/1095/2175.htm>.
13. The documents’ titles in Chinese are 义务教育语文课程标准 (2011年版), 普通高中语文课程标准 (2017年版) and 部编本教材名著导读专栏. The book lists were accessed from https://www.sohu.com/a/366643882_503492#google_vignette.
14. The books are *Hamlet*, *The Little Prince*, *Les Misérables*, *One Hundred Years of Solitude*, *Jane Eyre*, *The Old Man and the Sea*, and *Robinson Crusoe*.
15. Lu Xun, 1881–1926, is a writer and literary critic who is regarded as a leading figure of modern Chinese literature.
16. Lao She, 1899–1966, is a famous Chinese novelist and dramatist.
17. Rabindranath Tagore, 1861–1941, is a Bengali polymath who is most known for his poetry in China.
18. Jin Yong, 1924–2018, is famous for his Chinese WuXia fiction which combines martial arts, chivalry, and adventure.
19. Eileen Chang, 1920–95, is famous for her romance novels.
20. The machine translations are of good quality as the original tags are mostly short and straightforward.
21. The data and screenshots were accessed from the web pages <https://book.douban.com/subject/4722180/> (on Douban) and https://www.goodreads.com/book/show/4894.Who_Moved_My_Cheese (on Goodreads), on 6 May 2024.
22. The data was accessed from the web pages <https://www.goodreads.com/book/show/3961415-a-global-history> (on Goodreads) and <https://book.douban.com/subject/10583099/> (on Douban), on 18 May 2024.
23. The “jensenshannon” measurement was imported from the SciPy Python library, accessed from the webpage <https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.jensenshannon.html>, on 1 May 2025.

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